

◆ AWS CLOUD PRACTITIONER BOOTCAMP

WEEK 3 — SESSION 7

GITHUB LAB — NETWORKING FUNDAMENTALS

◆ LAB TITLE

AWS Networking Fundamentals Documentation Lab

◆ LAB OBJECTIVE

Students will:

- ✓ Understand Amazon VPC fundamentals
 - ✓ Differentiate public vs private subnets
 - ✓ Explain Internet Gateway functionality
 - ✓ Document AWS networking concepts professionally
 - ✓ Build visible proof-of-work on GitHub
-

◆ REQUIRED GITHUB STRUCTURE

Students must create/update:

```
aws-cloud-practitioner-bootcamp/  
├── Week-3-Networking/  
│   ├── vpc-basics.md  
│   ├── subnet-design.md  
│   ├── internet-gateway.md  
│   └── screenshots/
```

◆ STEP 1 — CREATE THE FOLDER

Inside the GitHub repository:

```
Week-3-Networking
```

◆ STEP 2 — CREATE `vpc-basics.md`

REQUIRED CONTENT

Students must explain:

1. What is Amazon VPC?

Include:

- Definition
 - Purpose
 - Why AWS uses VPCs
-

2. Why Is VPC Important?

Students should explain:

- Isolation
 - Security
 - Architecture control
-

3. Real-World Example

Example:

- Banking platform
 - Company internal network
 - Enterprise infrastructure
-



SAMPLE STRUCTURE

```
# Amazon VPC Basics
```

```
## What Is Amazon VPC?
```

```
Amazon VPC (Virtual Private Cloud) allows organizations to create isolated cloud networks inside AWS.
```

```
## Why VPC Matters
```

```
VPCs provide:
```

- ```
- Security
- Isolation
- Network control
- Traffic management
```

```
Real-World Example
```

```
A bank can deploy:
```

- ```
- web servers  
- databases  
- applications
```

inside a secure VPC environment.

◆ STEP 3 — CREATE `subnet-design.md`

REQUIRED CONTENT

Students must explain:

Public Subnet

Include:

- Internet access
 - Web servers
 - Load balancers
-

Private Subnet

Include:

- Databases
 - Internal systems
 - Security protection
-

Comparison Table

Public Subnet	Private Subnet
Internet accessible	No direct internet access
Hosts web apps	Hosts databases
Higher exposure	More secure

SAMPLE STRUCTURE

```
# Public vs Private Subnets
```

```
## Public Subnet
```

```
A public subnet allows resources to access the internet.
```

```
Examples:
```

- ```
- Web servers
- Load balancers
```

## Private Subnet

A private subnet protects sensitive systems from direct internet exposure.

Examples:

- Databases
- Internal applications

## Security Insight

Sensitive systems should remain private whenever possible.

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## ◆ STEP 4 — CREATE `internet-gateway.md`

### REQUIRED CONTENT

Students must explain:

- ✓ What an Internet Gateway is
  - ✓ Why it is required
  - ✓ How traffic flows
  - ✓ What happens without an IGW
- 

### SAMPLE STRUCTURE

# Internet Gateway

An Internet Gateway (IGW) allows communication between a VPC and the internet.

## Why It Matters

Without an IGW:

- users cannot access applications
- EC2 instances cannot reach the internet

## Example Traffic Flow

User → Internet → IGW → Public Subnet → EC2

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## ◆ STEP 5 — SCREENSHOTS (MANDATORY)

Inside:

screenshots/

Students upload:

- ✓ AWS VPC dashboard
  - ✓ Subnet page
  - ✓ Route table page
  - ✓ Internet Gateway page
- 

## ◆ STEP 6 — PUSH TO GITHUB

### COMMANDS

```
git add .
git commit -m "Completed Week 3 Networking Lab"
git push
```

---

## ◆ FINAL DELIVERABLE CHECKLIST

| Requirement              | Status |
|--------------------------|--------|
| Week-3-Networking folder | ✓      |
| vpc-basics.md            | ✓      |
| subnet-design.md         | ✓      |
| internet-gateway.md      | ✓      |
| Screenshots uploaded     | ✓      |
| GitHub push completed    | ✓      |

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## ◆ GOVERNANCE INSIGHT

Students must understand:

Networking is not just connectivity.

It controls:

- ✓ security
  - ✓ isolation
  - ✓ exposure
  - ✓ traffic flow
  - ✓ architecture quality
-

## ◆ BONUS CHALLENGE (HIGH-VALUE)

Students explain:

Scenario:

A database was placed in a public subnet.

Questions:

1. Why is this dangerous?
2. What risks exist?
3. How should the architecture be improved?

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## ◆ LAB OUTCOME

After completing this lab, students will:

- ✓ Understand AWS network structure
- ✓ Explain subnet segmentation
- ✓ Understand Internet Gateway functionality
- ✓ Build cloud documentation discipline
- ✓ Develop architecture thinking
- ✓ Strengthen GitHub portfolio visibility

